

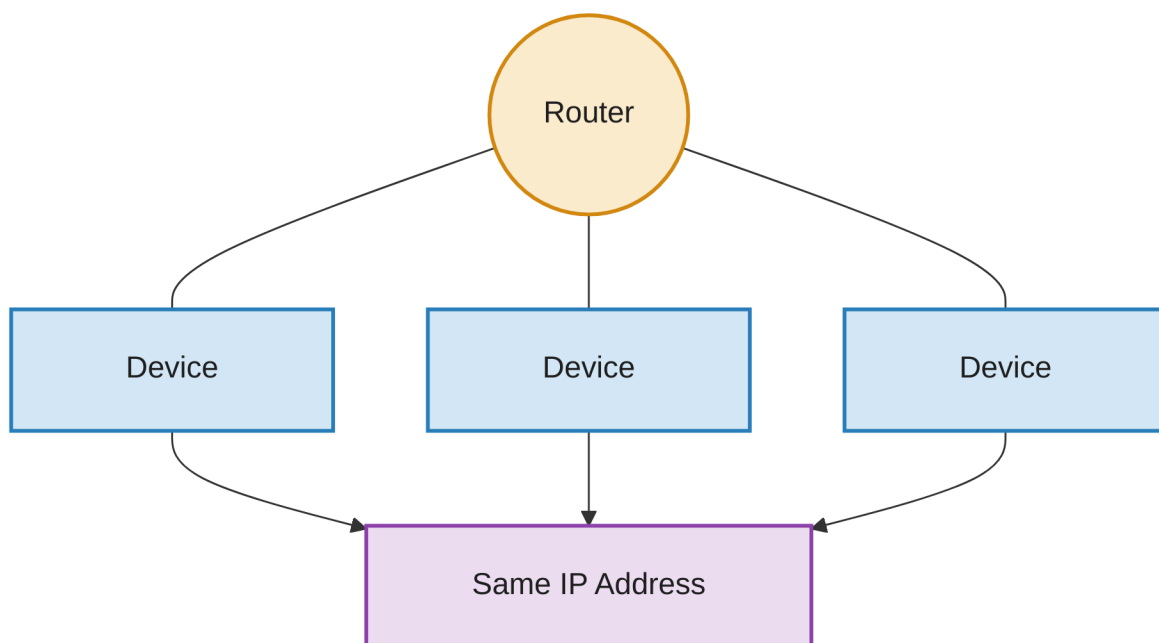
Lecture 5: Network Topology + VPN

Networks are organized at different scales — LAN, MAN, and WAN — and the devices within them can be connected using different physical layouts called topologies. VPNs build on top of this by changing how a device's traffic appears to the outside world.

Types of Network

1. LAN (Local Area Network)

- ♦ Used in small areas, such as a home, office, or a single building.



- ♦ All devices behind the same router share the same **public IP address** when seen from outside the network, while each device internally has its own **private IP address**.
- ♦ On a phone or laptop, only the public IP address is visible to the outside world — it represents the entire network as one unit.
- ♦ The router keeps track of which internal device made each request, so it can correctly route incoming responses back to the right device — this is done using NAT (Network Address Translation).

Private IP Address

A **Private IP Address** is reserved for use only within a private or local network — it is not visible to the wider internet.

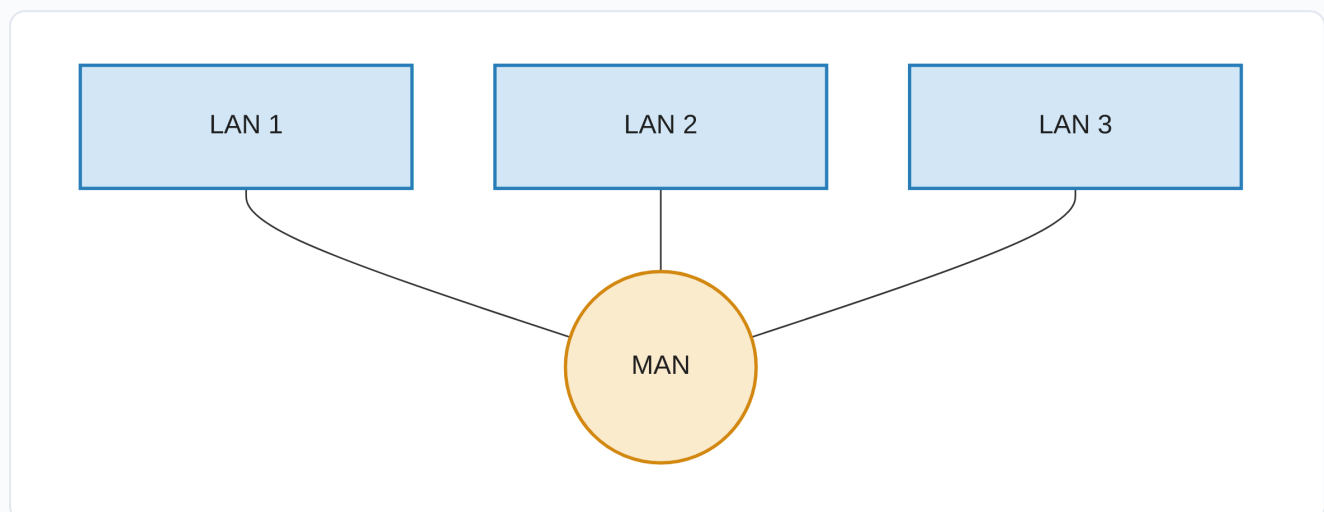
Private addresses were created because IPv4 has a limited number of addresses (~400 crore total) — reusing private ranges internally means the same private addresses can be reused across millions of different LANs worldwide without conflict.

Private IP Address Ranges:

CLASS	PRIVATE RANGE
Class A	10.0.0.0 – 10.255.255.255
Class B	172.16.0.0 – 172.31.255.255
Class C	192.168.0.0 – 192.168.255.255

2. MAN (Metropolitan Area Network)

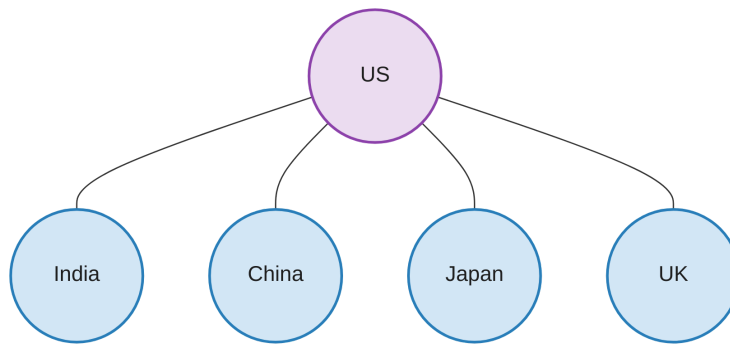
A MAN connects **multiple LANs** together.



- ♦ A MAN is essentially the connection of multiple LANs.
- ♦ Since each LAN is a separate private network, the **same private IP address can be reused** across different LANs (e.g. LAN 1 and LAN 2 can both have a device at `10.0.0.2`) without any conflict, because they are isolated from each other.

3. WAN (Wide Area Network)

A WAN connects networks across **large distances** — typically country to country.



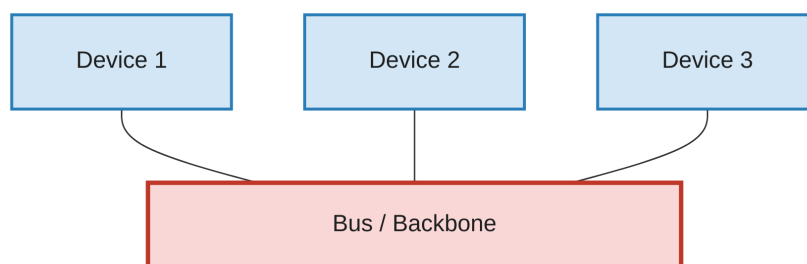
How does data travel across the world? Largely via **undersea cables** — the fastest method for country-to-country data transfer.

Types of Topology

A **topology** describes how devices in a network are physically or logically arranged and connected.

1. Bus Topology

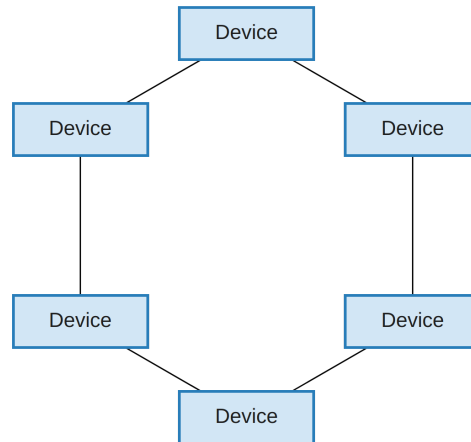
All devices connect to a single central cable, known as the bus or backbone. Data travels in both directions along the bus.



ADVANTAGES	DISADVANTAGES
Easy to set up and extend	If the main cable fails, the entire network goes down
Requires less cable than other topologies	Performance degrades as more devices are added

2. Ring Topology

Each device connects to **two other devices**, forming a circular pathway. Data travels in one direction (or both, in a dual ring).

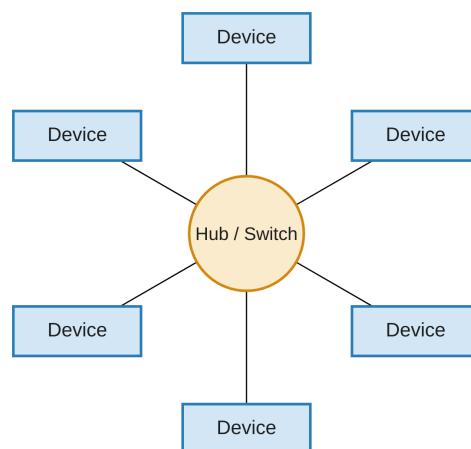


ADVANTAGES	DISADVANTAGES
Data packets travel at high speed	If one connection fails in a single ring, it can disrupt the entire network
Easy to identify faults and isolate them	Adding or removing devices can be disruptive

With a **single fault**, a basic ring can sometimes still route data the other way around; with **two or more faults**, the ring typically breaks and stops working entirely.

3. Star Topology

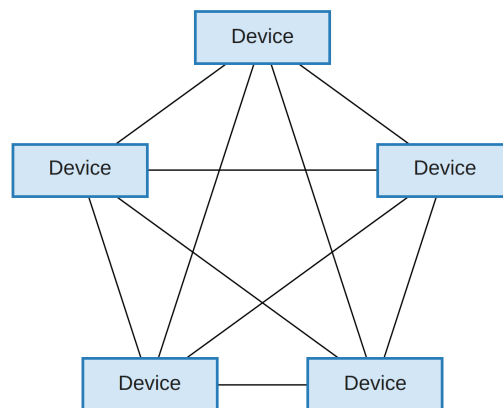
All devices connect to a **central hub or switch**. Data passes through the hub/switch before reaching its destination.



ADVANTAGES	DISADVANTAGES
If one connection fails, it does not affect the rest of the network	If the central hub fails, the entire network goes down
Easy to add or remove devices	Requires more cable than Bus topology

4. Mesh Topology

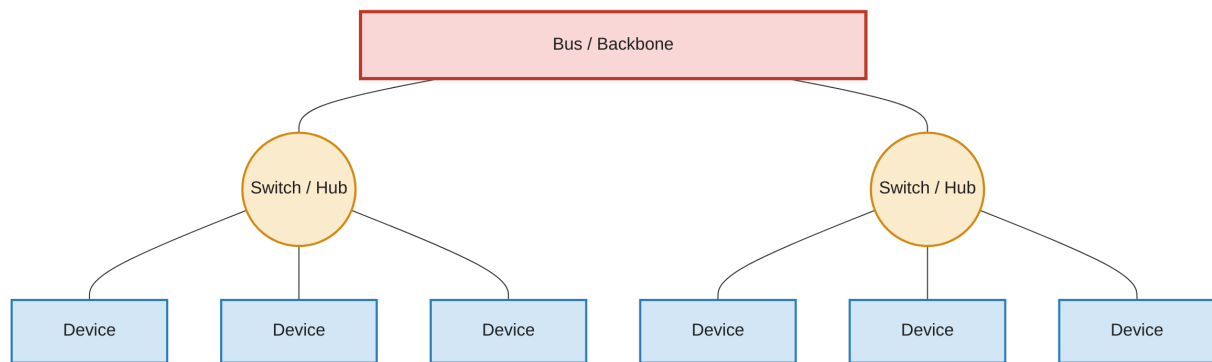
Every device connects to **every other device**, either **fully** (full mesh) or **partially** (partial mesh).



ADVANTAGES	DISADVANTAGES
Highly reliable — if one connection fails, others take over	Expensive to implement due to the high number of wires needed
Offers redundancy and multiple pathways for data	Difficult to set up and manage; adding new devices is also difficult

5. Hybrid Topology

A **combination of two or more different topologies** — for example, Star-Bus or Star-Ring.



ADVANTAGES	DISADVANTAGES
Flexible and scalable	Complex to design and implement
Can be designed to suit specific network requirements	Can be expensive, depending on the mix of topologies used

Hybrid topologies of this kind are typically seen **only within a LAN**.

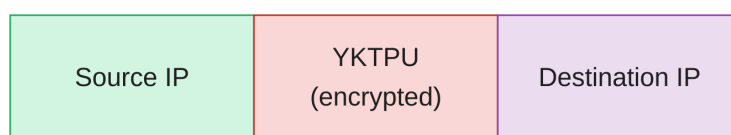
Encryption and Online Privacy

- ♦ Data sent over the internet is **encrypted** to prevent hackers from reading it and to protect user privacy.

Example (conceptual):

Original: DSA by Coder Army
 Encrypted (scrambled): YKTPU

- ♦ Data is encrypted using cryptography.
- ♦ It is **decrypted only when it reaches the server**.



The middle portion shown here represents what the original data looks like once scrambled through an encryption algorithm — completely unreadable to anyone intercepting it, without the correct decryption key.

HTTP vs HTTPS

PROTOCOL	DATA ENCRYPTION
HTTP	Data is not encrypted
HTTPS	Data is encrypted

Incognito Mode — A Common Misconception

*Incognito Mode only means your **local browsing history** won't be stored on your device. Your ISP can still see **which websites** you visit, because the destination IP address is visible in every request — it just doesn't know the specific content/videos/pages within that site.*

Privacy is, to a large extent, a myth — because your ISP still knows which websites you visit, even in incognito mode.

How Are Websites Banned in a Country?

When a government wants to block a website, it instructs ISPs to enforce the ban. ISPs implement this by blocking the site's **IP address** or **domain name**, so requests to that site are refused before they ever reach it.

VPN (Virtual Private Network)

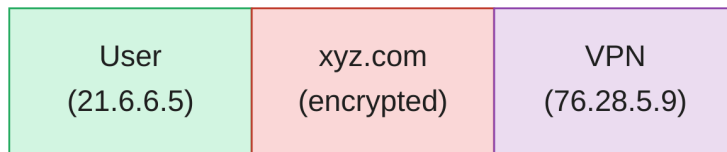
A **VPN** routes traffic through an intermediate server, which changes your visible IP address.



The VPN server acts as an intermediary between you and the destination — the destination (and your ISP) sees the **VPN's IP address**, not your original one.

How VPN Helps Access Restricted Websites

If your ISP blocks a certain website, the VPN reroutes the request like this:



- ♦ The ISP only sees that you are contacting the **VPN**, not the actual banned website — so it **allows** the request through.
- ♦ The VPN then goes to the banned website **on your behalf**, retrieves the content, and relays it back to you (encrypted along the way).



Real-World Example: TikTok vs CapCut in India

SITE TYPE	VPN ACCESS IN INDIA
TikTok	Cannot be accessed even via VPN — blocked at a level where the government actively prevents VPN traffic from connecting to it
CapCut	Can be accessed via VPN — blocking for it is not enforced as strictly

*This illustrates that **blocking enforcement varies** depending on the type of content and the relevant policy decisions.*

Summary

- ♦ Networks are organized by scale: LAN (small area, single router, private and public IP via NAT), MAN (multiple connected LANs), and WAN (country-to-country, largely via undersea cables).
- ♦ Private IP addresses (reserved ranges in Class A, B, and C) let internal devices share IPs safely, since IPv4's limited address pool can't give every device a unique public IP.
- ♦ NAT (Network Address Translation) lets a router track which internal device made a request — using both the IP mapping and the device's MAC address — so it can correctly deliver responses to the right device.
- ♦ Topologies describe how devices are physically or logically connected: Bus (single shared cable, cheap but a single point of failure), Ring (circular connections, fast but vulnerable to multiple faults), Star (central hub, isolates device faults but depends on the hub), Mesh (fully interconnected, highly reliable but expensive), and Hybrid (a combination, offering flexibility at the cost of added complexity).
- ♦ Data in transit is encrypted using cryptography, so it appears as unreadable scrambled text to anyone intercepting it, and is only decrypted at the destination server — this is the basis of HTTPS versus HTTP.
- ♦ Incognito Mode only stops browsing history from being saved locally — the ISP can still see which websites are visited, because of the destination IP address, making full privacy largely a myth.
- ♦ Governments enforce website bans by having ISPs block a site's IP address or domain name.
- ♦ A VPN reroutes traffic through an intermediate server, changing the visible IP address so the ISP only sees a request to the VPN, not the actual restricted destination site.
- ♦ Enforcement of VPN-based blocking varies by content type — some sites remain blocked even through a VPN, while others are accessible.